

A Permissionless, Ownerless Supply-Chain Assembly: Why TradeLens Failed and the Consumer-Rooted Architecture That Succeeds It

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Abstract

TradeLens (2018–2023), the IBM–Maersk container-shipping consortium that aimed to produce a tamper-proof shared record of every container’s journey, shut down with only one major non-Maersk carrier on board. The shutdown was not a software failure. It was a governance failure visible from the consortium’s structural shape: a competitor-controlled platform, requiring participation from carriers who would not ratify a competitor’s gate-keeping over an industry resource, with high integration cost and no settlement guarantee binding parties to the arrangement. We argue the failure was structural, predictable from the shape of the architecture, and not specific to TradeLens.

TradeLens addressed the maritime portion of a longer chain. The architectural alternative we develop reaches further: from factory to consumer. We present a permissionless, ownerless supply-chain assembly built on the bonded settlement primitive of the companion mechanism paper. The assembly has no consortium, no foundation, no admin function, no upgrade path, and no central party who can be ratified or refused. The *final consumer* is the **rootBuyer** of the supply-chain process, committing directly to every party that contributes to the goods arriving at the consumer’s door — factory, packaging, origin logistics, port authorities at origin and destination, ocean carrier, customs broker, customs authority (for import duties), domestic logistics, state government (for sales tax), and any other resource provider the goods draw on. Carriers who want to participate compose; carriers who do not do not, with no consequence for either decision. Bonded commitments at each commit lock skin in the game proportional to the cumulative process value; atomic resolution by the consumer settles the whole process on receipt of the goods. The firm-shape intermediaries the conventional supply chain inserts at every tier (shipper-of-record, distributor, wholesaler, retailer) become structurally unnecessary because the consumer’s bonded commits reach the operational sellers directly. The schemas the assembly draws on are largely existing infrastructure (commerce, geo, handoff, jurisdiction, fulfilment, GHG family, proximity); two new schemas (container-seal, Incoterms-2020-as-anchor) extend the protocol’s surface in ways that respect the extension-doctrine. We also treat the bill-of-lading transferability question through the cancellable-seller / counter-process pattern that expresses transferability at the protocol layer without weakening the kernel’s invariants.

The paper is in industrial-engineering register: the contribution is the assembly DAG and its operational properties, not novel mechanism-design results. Why TradeLens failed and what its successor looks like architecturally are answerable as supply-chain coordination questions.

Keywords: container shipping, supply-chain coordination, bills of lading, Incoterms, MLETR, transferable records, process modeling

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1 Introduction

Container shipping is a coordination problem of unusual scale: a single 20,000-TEU vessel carries cargo from thousands of shippers, moves through multiple ports under multiple jurisdictions, transfers to multiple consignees, and is documented by a stack of bills of lading, customs filings, GHG attestations, and operational handoffs that no single party originates and no single party owns. The information problem is real — duplicated paper-based records, opaque vessel-position data, contested seal-integrity claims — and the industry has been searching for a digital solution for two decades.

TradeLens, the consortium platform jointly developed by IBM and Maersk between 2018 and 2023, was the most ambitious attempt at that solution. Its approach was a permissioned blockchain on which participants would publish operational records that all parties to a shipment could read. The platform shut down in early 2023 after failing to attract more than one major non-Maersk carrier (CMA CGM) into participation; MSC, Hapag-Lloyd, ONE, and Evergreen all declined. The platform was technically functional; the consortium was not.

Reading the failure structurally. The standard reading of TradeLens’s failure focuses on adoption dynamics: the network-effects argument that any shared-ledger platform requires a critical-mass coalition before it is useful. This is not wrong, but it is incomplete. The critical-mass argument explains why TradeLens *failed to grow*; it does not explain why the carriers who declined to join made what was, from their perspective, a rational decision. The carriers’ decision was not about technology adoption. It was about ratifying a competitor’s control of an industry resource, which a competitor-controlled platform necessarily asks them to do.

The *governance* structure of the platform was the binding constraint, not the network effects. A competitor-controlled platform requires the other competitors to accept the gatekeeping authority of the controlling competitor; the other competitors have correctly read this as conceding strategic position and have declined. The network-effects collapse follows from the governance choice; it does not cause it. A platform with neutral governance might have faced its own difficulties, but it would not have faced the specific difficulty TradeLens did.

The architectural alternative. We argue that the structural fix is not better governance but no governance at the platform layer at all. A permissionless, ownerless settlement primitive — the bonded commitment of the companion mechanism paper — removes the governance question from the architecture. There is no consortium to join; there is a public protocol that any wallet can compose with. There is no gatekeeping authority to refuse; carriers participate by signing commitments and decline by not signing. The network-effects question becomes more manageable because the platform-as-veto problem disappears: a competitor’s participation does not require the competitor’s consent to a competitor’s gatekeeping.

The assembly described in this paper is what supply-chain coordination looks like under such a substrate. TradeLens addressed the maritime portion of the chain — carrier-to-carrier and carrier-to-port records of cargo movement; the bonded architecture reaches further. It admits the *full chain from factory to consumer* under a single settlement primitive, with the final consumer as the **rootBuyer** of the supply-chain process and every operational party (factory, packaging, origin logistics, port authorities, ocean carrier, customs broker, customs authority for tariffs, domestic logistics, state government for sales tax, and any other resource provider the goods draw on) as a seller in that process. The firm-shape intermediaries of the conventional chain — shipper-of-record, distributor, wholesaler, retailer — become structurally unnecessary because the consumer’s bonded commits reach the operational sellers directly. It is not a TradeLens replacement in the sense of a drop-in alternative product; it is a re-architecture of the supply-chain coordination problem in which the firm-tiered chain is replaced by a consumer-rooted bonded process.

Paper organization. Section 2 reviews the TradeLens history and the structural reading of its failure (the maritime case). Section 3 summarizes the settlement primitive in the register the present paper uses. Section 4 develops the consumer-rooted assembly: the organizational DAG (factory to consumer), the schema-attested operational handoffs along it, and the kernel commit structure that backs it (consumer as `rootBuyer`, every commit consumer-to-seller, all in one process). Section 5 works out the bond posture and cohort-compensation mechanics across the resource providers, with a worked example including realistic numbers, tariffs, and sales tax. Section 6 treats the schema requirements: most existing infrastructure, two new schemas (container-seal-v1 and Incoterms-2020-as-anchor), and the deliberate non-inclusion of transferable-BoL semantics in the kernel-layer schema set. Section 7 treats bill-of-lading transferability as a protocol-layer composition pattern (CancellableSeller wrapper + counter-process). Section 8 compares the architecture to CargoX, TradeTrust, MLETR, and TradeLens itself. Section 9 states the scope conditions and what the architecture deliberately does not address. Section 10 concludes.

2 What TradeLens Was and Why It Failed

We treat the failure structurally rather than narratively.

What TradeLens was. TradeLens was a permissioned blockchain platform for global container shipping, jointly developed by IBM and Maersk and operational from 2018 to 2023. The platform’s stated goal was a shared, tamper-proof record of every container’s journey — bills of lading, customs filings, port handoffs, vessel positions — accessible to all parties in a shipment.

The technical architecture was Hyperledger-Fabric-based: a permissioned ledger where participating carriers, ports, and customs authorities could publish operational records. The platform’s value proposition rested on the assumption that all major ocean carriers would join, producing a single ledger that shippers could rely on as comprehensive cross-carrier reference. Jensen et al. [2019] and Sarker et al. [2021] document the platform’s value proposition and use-case design as it stood mid-deployment; both papers predate the shutdown and frame the platform around adoption potential rather than around the governance refusal we read as the binding constraint below. The governance-was-binding reading is original to this paper; Jensen and Sarker supply the value-proposition baseline against which the reading is offered.

The adoption shape. By 2022, only CMA CGM (a Maersk competitor) had joined as a major non-Maersk carrier. MSC (the largest carrier by capacity), Hapag-Lloyd, Ocean Network Express (ONE), and Evergreen had all declined to participate. In November 2022, IBM and Maersk announced the platform’s shutdown; production ended in Q1 2023 [Lloyd’s Loadstar, 2022].

Why the carriers declined. The standard adoption-dynamics reading explains the shutdown through a familiar mechanism: shippers want a single ledger covering 90%+ of their cargo volume; without MSC and Hapag-Lloyd the data was incomplete; without complete data, shippers wouldn’t pay for the platform; without revenue, carriers had no incremental reason to join; the equilibrium was reflexive non-adoption. This is correct as far as it goes.

The deeper reading is that the non-Maersk carriers correctly identified a strategic problem with the platform’s governance shape. A platform controlled by Maersk, with IBM as the technical operator and Maersk as the principal customer, is structurally a Maersk asset that other carriers are being asked to provision with their operational data. The asset’s governance accountability runs to Maersk in any contested case. Whatever the platform’s formal governance arrangements, the controlling-competitor dynamic is the binding constraint on competitors’ participation.

The argument is not that Maersk acted in bad faith or that the formal governance was inadequate. The argument is that the platform’s shape *is* the problem: a competitor-controlled infrastructure requires other competitors to ratify the controller’s strategic position relative to the industry, and there is no formal-governance fix that dissolves this requirement. MSC and Hapag-Lloyd are not joining a Maersk-controlled platform regardless of how its governance is configured.

The general lesson. TradeLens’s failure is not specific to IBM, to Maersk, or to shipping. It is the recurring pattern of consortium-governance infrastructure: any platform whose governance accountability runs to a participant of the industry the platform is meant to serve will face structural participation refusal from that participant’s competitors. The fix is not better consortium governance; the fix is no consortium at the platform layer.

Several adjacent industries have run the same experiment with the same outcome: SWIFT and its alternatives in payments, various consortium attempts at securities settlement, healthcare-records consortia. The pattern is robust. The structural alternative is permissionless, ownerless infrastructure at the platform layer, with whatever industry-specific arrangements the participants want to compose on top.

3 The Settlement Primitive

The bonded primitive is developed formally in the companion mechanism paper, with the deployed Solidity implementation verified through the companion implementation paper’s multi-method stack (TLA⁺, symbolic execution, property-based fuzzing, Certora specifications). We summarize the features the present argument uses.

Asymmetric bonding (Mechanism 1). For a transaction with payment $P > 0$ and cumulative upstream value $G \geq P$, the buyer locks $2P$ and the seller locks $2G$. Cooperation is the unique profile surviving iterated elimination of weakly dominated strategies. The mechanism scales to N -party process trees through progressive collateralization: each seller bonds against the cumulative value of all upstream commitments, producing a mesh of independently secured bilateral edges, each carrying its own equilibrium.

Buyer dominance with atomic resolution (Mechanism 2). Only the root buyer can trigger resolution; resolution settles all active orders in the process simultaneously or not at all. The all-or-nothing rule induces a weakest-link subgame among sellers, producing cooperation pressure of magnitude $P_i + 2G_i$ on every other seller from each seller’s perspective, without explicit communication, governance, or managerial layer.

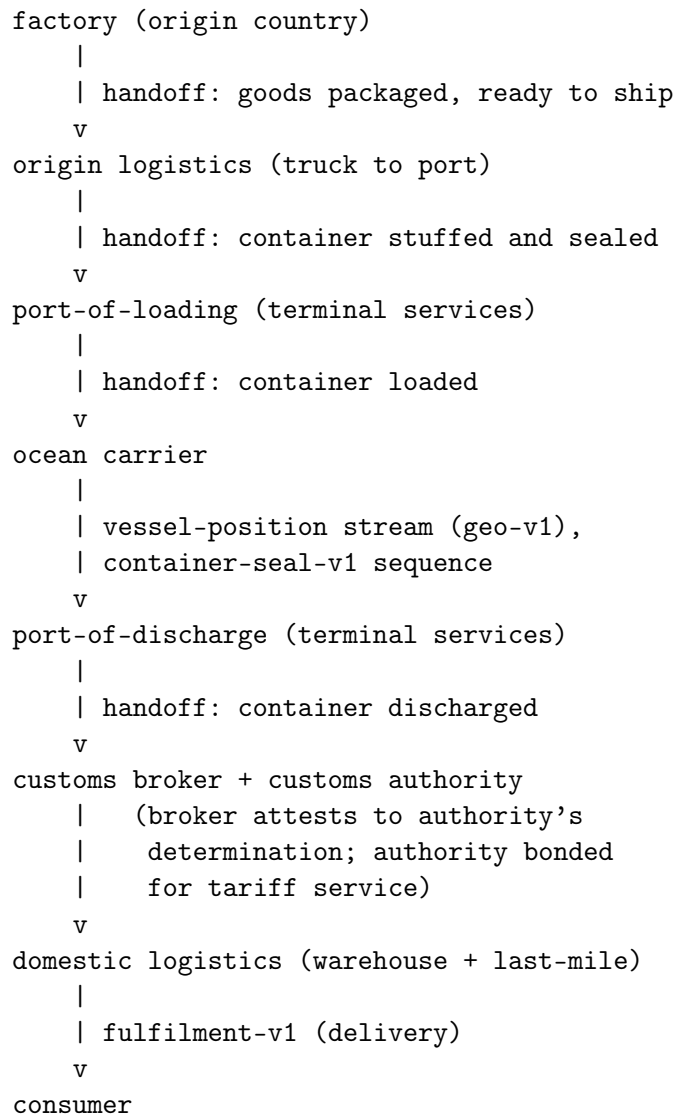
What this implies for shipping coordination. Mechanism 1 prices each leg’s enforcement in proportional bond posture without requiring any consortium or platform to underwrite it. Mechanism 2 enforces the multi-leg coordination at resolution without requiring any party to govern the multi-leg arrangement. The two mechanisms together supply the coordination function that TradeLens attempted to supply through a consortium-governed shared ledger; they supply it without needing the consortium.

4 The Assembly

We now describe the assembly of bonded commitments that re-architects the supply chain as a permissionless, ownerless coordination protocol. The assembly has two layers that should be kept clearly separate: the *organizational DAG* (how the goods physically flow from factory to

consumer, documented via schema attestations) and the *kernel commit structure* (who commits to whom on-chain, governed by the kernel’s `buyer = rootBuyer` rule). The kernel is DAG-blind; the schema layer is DAG-aware. The two layers co-exist in one process.

The organizational DAG. The supply chain is a sequence of operational handoffs from the factory to the consumer. We sketch the canonical international-DTC flow:



Variations on the structure (factory-direct shipment without a warehouse intermediary, inter-modal rail substituted for ocean, etc.) instantiate the same pattern with the operational divisions shifted to reflect the route. Every handoff along this DAG is documented via a schema-typed attestation: `handoff-v1` for cargo transfers, `container-seal-v1` for seal-state changes, `geo-v1` for vessel-position updates, `jurisdiction-v1` for customs clearance, `fulfilment-v1` for delivery confirmation. None of these attestations is a kernel commit; they are evidentiary records that document the organizational reality of the goods’ flow.

The kernel commit structure. The kernel commits, in contrast, are consumer-rooted: the kernel enforces `buyer = rootBuyer` in every order in a process, so every kernel commit in the supply-chain process has the consumer as buyer. The commits document who the consumer pays and how much: `consumer → factory`, `consumer → origin logistics`, `consumer → origin port authority`, `consumer → ocean carrier`, `consumer → destination port authority`, `consumer → customs`

broker, consumer → customs authority (for the import tariff), consumer → domestic logistics, consumer → state government (for sales tax), and any other resource provider the goods draw on. The commits are independent of who hands the goods to whom in the organizational DAG above. The consumer alone calls `resolveProcess` on receipt, settling the process atomically.

Why this is not a chain of B2B transactions. In the conventional supply chain the same goods are sold and resold at every tier — factory to wholesaler, wholesaler to distributor, distributor to retailer, retailer to consumer — and each tier extracts margin in exchange for absorbing the coordination cost between adjacent tiers. The bonded primitive collapses that coordination cost: the consumer’s bonded commits reach the operational sellers directly, with no margin-extracting tier in between. The factory is paid by the consumer, not by a wholesaler. The ocean carrier is paid by the consumer, not by a shipper-of-record. The firm-shape parties of the conventional chain (shipper-of-record, distributor, wholesaler, retailer) become structurally unnecessary at the kernel layer; off-chain roles such as discovery, curation, brand-as-quality-signal, or returns handling can still be filled by parties who provide those specific services, but they cannot insert themselves as kernel intermediaries between the consumer and the operational sellers.

Per-commit schema clauses. Each kernel commit carries the schema clauses relevant to the seller’s role.

- *consumer → factory*: `commerce-v1` payment, `handoff-v1` on goods packaged for shipment, optional `ghg-protocol-v1` disclosure for the manufacturing emissions.
- *consumer → origin logistics*: `commerce-v1` payment, `handoff-v1` on cargo receipt and on container stuffing, `container-seal-v1` applied at the factory dock or origin warehouse.
- *consumer → origin port authority* and *consumer → destination port authority*: `commerce-v1` payment for terminal handling and gate access, `handoff-v1` at port-side transfers, and `container-seal-v1` inspected-intact attestation at the port handoff.
- *consumer → ocean carrier*: `commerce-v1` payment, an `incoterms-2020-v1` clause anchoring the shipment term and named place, `handoff-v1` on cargo transfer to the vessel and at discharge, `container-seal-v1` sealing events through the voyage, `geo-v1` vessel-position attestations on the voyage cadence, and `ghg-protocol-v1` disclosure for the voyage emissions.
- *consumer → customs broker*: `commerce-v1` payment plus `jurisdiction-v1` attestation recording the customs authority’s determination.
- *consumer → customs authority (import duty / tariff)*: `commerce-v1` payment for the import-duty obligation. The customs authority is a bonded seller of border-services in this commit and a sovereign decisional authority on the broker’s parallel attestation; the two roles are distinct but compose in one process.
- *consumer → domestic logistics*: `commerce-v1` payment plus `fulfilment-v1` attestation at delivery.
- *consumer → state government (sales tax)*: `commerce-v1` payment for the sales-tax obligation. The state government is a bonded seller of public services funded through the sales tax, on the same composition pattern as the customs authority.

Operational sub-procurement as separate processes. A factory’s procurement of materials, labor, and energy; a carrier’s procurement of fuel, crew, and maintenance; a domestic-logistics provider’s procurement of warehouse space and delivery vehicles — each of these is a separate process under the operational party as its own `rootBuyer`. The kernel does not link those processes to the consumer’s. Cross-process coordination is achieved through within-process cohort-pressure on each side and through off-chain reconciliation, not through cross-process atomic resolution.

5 Bond Posture and Cascade Mechanics

The architecture’s structural contribution is that each leg’s risk is priced and propagated by the bonding rule of the kernel, without consortium intermediation.

Bond posture per commit. The kernel maintains a single process-wide cumulative-value accumulator G that grows monotonically as each commit lands ($G \leftarrow G + P_i$). At every commit the consumer (as `rootBuyer`) posts $2P_i$ for that specific payment, and the seller posts $2G_{\text{at commit time}}$ — the full cumulative process value at the moment they commit. Bond posture is asymmetric: the consumer’s per-commit bond scales with the immediate payment, while each seller’s bond scales with the cumulative value of every upstream commit in the process. Sellers committing later post correspondingly larger bonds; the last seller bonds against the full retail value of the goods.

A bond-posture worked example. Consider a stylized \$200 imported consumer good (a piece of consumer electronics manufactured in an exporting country and delivered to a consumer in an importing country with a 25% import duty assessed on the CIF landed cost and a 7.5% sales tax assessed on the pre-tax retail price) under one consumer-rooted process. The table below traces one illustrative commit sequence.

Commit (in sequence)	P_i	G after commit	Consumer bond $2P_i$	Seller bond $2G$
Root: consumer → factory (FOB)	\$85	\$85	\$170	\$170
Sub: consumer → origin logistics	\$5	\$90	\$10	\$180
Sub: consumer → origin port authority	\$5	\$95	\$10	\$190
Sub: consumer → ocean carrier	\$15	\$110	\$30	\$220
Sub: consumer → destination port authority	\$5	\$115	\$10	\$230
Sub: consumer → customs broker	\$3	\$118	\$6	\$236
Sub: consumer → customs authority (import duty)	\$29	\$147	\$58	\$294
Sub: consumer → domestic logistics	\$39	\$186	\$78	\$372
Sub: consumer → state government (sales tax)	\$14	\$200	\$28	\$400
Total bond locked across the process	—	—	\$400	\$2,292

Table 1: Stylized per-commit bond posture for a \$200 imported consumer good under one consumer-rooted process. The import duty (\$29) is 25% of the CIF landed cost (\$115 = \$85 + \$5 + \$5 + \$15 + \$5, the cumulative process value at point of arrival at the destination port), not of the retail price. The sales tax (\$14) is 7.5% of the pre-tax retail (\$186 = \$200 − \$14). G is the process-wide cumulative-value accumulator. The consumer’s bond at each commit scales with the immediate payment to that seller; each seller’s bond scales with the cumulative process value at the moment that seller commits. The customs authority and the state government are bonded sellers in their respective commits, on the same composition pattern as the federal aviation system in the companion scheduled-service paper. Per-resource payments are illustrative; absolute magnitudes vary with the goods, route, tariff schedule, and tax jurisdiction.

The asymmetry between consumer-side and seller-cohort bonds is visible: the consumer’s aggregate locked bond is \$400 ($= 2 \times P_{\text{retail}}$, the standard buyer posture), while the seller cohort’s aggregate locked bond is \$2,292, nearly six times the consumer’s. The seller cohort’s bond is a multiple of the retail value because each seller bonds against the cumulative G that has grown across upstream commits.

Slip propagation as cohort-compensation pressure. When a slip occurs — a defective unit shipped, a port congestion delay, a customs hold, a damaged delivery — the kernel’s atomic-resolution rule means *every* seller’s bond in the consumer’s process is at risk: the consumer alone resolves, and the consumer can withhold resolution while the seller cohort negotiates compensation. The unit whose performance failed bears the larger share of cohort-internal compensation; sellers whose service was unaffected contribute less or not at all. Compensation flows directly from the seller cohort to the consumer before any external mechanism (specialized mechanism contracts, dispute forums, insurance, regulators) is engaged. The mechanism is the same one developed for scheduled passenger service in the companion industrial-engineering paper: buyer dominance plus atomic resolution under bond pressure. The off-chain question of which operational unit caused the slip — factory defect, packaging fault, ocean-leg damage, customs inspection delay — is adjudicated by external forums with the bonded record as input, under the framework of the companion evidence-and-law paper.

Cohort pressure across the resource provider set. The architecture exhibits cohort pressure across the seller set in the consumer’s process. Each seller’s bond is at risk if any other seller’s slip propagates to a resolution-withholding by the consumer. Sellers therefore have direct economic interest in each other’s reliability: a factory whose defect rate forces repeat compensation negotiations loses repeat consumer business not because of reputation effects but because the cohort’s expected bond-loss exposure on goods from that factory is direct and measurable; a port that consistently produces cascading delays does likewise. The mechanism operates without a managerial layer; the bonded structure does the work that a consortium-governed shared ledger was meant to do through information transparency alone.

6 Schemas

The assembly draws mostly on existing schema infrastructure. Two new schemas are required: a container-seal attestation and an Incoterms-2020 anchor. The bill-of-lading transferability question is treated separately in Section 7.

Existing schemas. The assembly composes the following existing schemas. `figaro-commerce-v1` carries the per-leg payment commitment. `figaro-geo-v1` carries pickup / drop-off geohashes and vessel-position updates. `figaro-handoff-v1` carries each container handoff at transfer points. `figaro-jurisdiction-v1` carries the customs-clearance attestation. `figaro-courier-process-v1` carries the carrier’s internal sub-process tree (the multi-stop carrier process). `figaro-fulfilment-v1` carries the final delivery to consignee. `figaro-ghg-protocol-v1` (with sister schemas for ISO 14064, PAS 2050, EN 16258) carries per-leg emissions disclosure on every transport edge. `figaro-proximity-policy-v1` together with `figaro-proximity-proof-v1` support geofence enforcement at port handoffs where required.

The composition is non-trivial but draws on established infrastructure. No agent work is required for these schemas; they are already on chain and bound to validators.

New schema: `figaro-container-seal-v1`. Container seals are the integrity mechanism for ocean shipping. A uniquely-numbered seal is applied at the origin and broken only at customs or

at the destination consignee. Multiple parties (carrier, ports, customs, consignee) need a single canonical attestation of when the seal was applied, when it was inspected intact, and if it was breached.

The schema’s Layer A spec carries:

- `containerNumber` (string, ISO 6346 format)
- `sealNumber` (string)
- `event` (enum: applied, inspected_intact, transferred, breached, removed_by_customs)
- `inspectorAddress` (address-hex; the wallet making the attestation)
- `timestamp` (ISO 8601 datetime)
- `locationGeohash` (string, 8-character precision)
- `evidenceHash` (bytes-hex, optional content hash of a photo or inspection report)

The schema is the seed of a *container-integrity* family that could later cover seal-equivalents in other transport modes (rail container seals, air-freight tamper-evident packaging, etc.) without reauthoring the family. The `schemaId` binds permanently under the `SchemaRegistry`’s first-write-wins discipline.

New schema: figaro-incoterms-2020-v1. Incoterms 2020 (from the International Chamber of Commerce) are the international trade vocabulary for risk-transfer points and cost allocation across legs of a shipment: EXW, FCA, CPT, CIP, DAP, DPU, DDP, FAS, FOB, CFR, CIF [ICC, 2020, Ramberg, 2010]. The vocabulary is useful as cross-party shared reference, but the traditional Incoterms semantics come from a context without the bonded primitive’s invariants and may not all map cleanly. We treat this carefully because it is the canonical worked example of how traditional commercial vocabulary smuggles in assumptions the bonded architecture cannot honor.

Three structural mismatches deserve direct treatment.

Risk transfer is a Figaro-foreign concept. Incoterms encode a discretionary risk-transfer point: “risk transfers from seller to buyer at point X.” Under the bonded primitive, risk follows bond — the bonded party is at stake until they perform per the agreed clauses, and there is no separate *risk-transfer* concept distinct from the bonding rule. A clean port from Incoterms to bonded clauses requires reinterpreting “risk transfer at X” as “the seller’s `handoff-v1` attestation due at X discharges the seller’s bond exposure for the upstream segment.”

Cost-vs-risk separation collapses. Incoterms separate “who pays for transport” from “who bears risk during transport.” Under the bonded primitive, both follow from process structure: a party who bonds for a sub-process is the buyer of that sub-process, and risk is what the bonding rule already prices. Some Incoterms features (CIP/CIF’s seller-pays-insurance feature, for instance) require composition with a parallel insurance process rather than direct encoding in the delivery-clause specification.

Buyer dominance changes the resolution model. Incoterms assume good-faith resolution between parties; the bonded primitive encodes buyer dominance plus mutually-assured bond-loss as the equilibrium that produces good-faith resolution. Some Incoterms’ implicit dispute-resolution paths — which typically rely on local commercial-court adjudication — continue to function under the bonded architecture but at the off-chain forum the parties name in their agreement, not as an internal property of the term.

The right schema is therefore not “Incoterm as a traditional contract clause” but *Incoterm as a reference to a Figaro-native delivery-clause specification*. The schema anchors the term plus the named place; the term-to-delivery-clause mapping lives in the off-chain ICC publication

(the canonical text) and a runtime term-to-clause table that the assembly’s participants compose from. The validator contract enforces the anchor (term enum, named place string, place geohash for cross-reference with `geo-v1`) but does not encode per-term semantics in Solidity.

The Layer A spec carries:

- `term` (enum: EXW, FCA, CPT, CIP, DAP, DPU, DDP, FAS, FOB, CFR, CIF)
- `namedPlace` (string; every Incoterm requires a named place)
- `placeGeohash` (string, 8-character precision for cross-reference with `figaro-geo-v1`)

This split lets future Incoterms revisions ship as new schemaIds (`figaro-incoterms-2030-v1`, etc.) without mutating prior anchors. Per-term mapping verification — which terms map directly, which require composition with parallel processes (insurance, customs), and which do not transfer at all — is work the schema-author performs against the kernel code before the schema is registered. Likely outcomes from that verification are the following.

EXW, FCA, DAP, DPU, DDP, FAS, FOB probably map directly, each becoming a delivery-clause specification: `handoff-v1` attestation at the named place, with auxiliary clauses for customs (DDP) or unloading (DPU) where the term requires them.

CPT and CFR map structurally. The “seller pays carriage” feature becomes a separate sub-process where the seller is the buyer of carriage. The “risk transfers at first carrier” feature becomes the goods-commit’s delivery-clause specification.

CIP and CIF map similarly to CPT and CFR plus the “seller insures for buyer’s benefit” feature. The insurance assignment may not transfer directly — it likely requires composition with a parallel insurance process where the seller buys insurance from an external insurer naming the buyer as the beneficiary. The schema-author’s verification report identifies this as a composition requirement rather than as a refusal.

The Incoterms schema is therefore a clean illustration of the extension-doctrine principle that the protocol anchors shared references but does not import every contractual assumption that travels with traditional vocabulary. The vocabulary is useful; the assumptions are sometimes incompatible; the schema separates them.

Schemas the assembly does not include. The assembly does not include a multi-currency-cross-leg schema (would break same-unit comparability and the bonding equilibrium), a force-majeure-escape schema (would re-introduce the kind of unilateral exit the kernel forbids), or a transferability schema internal to `commerce-v1`. The transferability question is treated separately in Section 7 as a protocol-layer composition pattern.

7 Bill of Lading Transferability

The bill of lading is the carrier’s receipt of cargo, the contract of carriage, and (in the negotiable case) a document of title. The transferability question — whether the right to receive cargo at destination can be transferred mid-shipment from the original consignee to a new consignee — is doctrinally significant because container-shipping commerce often involves cargo sale in transit, and the canonical legal artifact for that sale is endorsement of the bill of lading.

The bonded primitive’s kernel does not support party substitution within a single bonded order. Three kernel invariants each independently rule it out: the buyer is fixed at root commitment and identical across every sub-order in the process; the parties to each commitment are fixed at the moment of commit; and the no-escape-hatches discipline forbids unilateral exit paths from a bonded state. A literal endorsement of the BoL during shipment would require any of these invariants to bend, and they do not bend.

The architectural direction we propose, and which we treat as *open architectural research* rather than as a deliverable production-ready pattern, is that BoL transferability is a *protocol-layer composition* not a kernel-layer property [BoL Research, 2026]. The candidate pattern is a *cancellable-seller wrapper plus counter-process*: the original buyer, who wants to transfer the right to the cargo mid-shipment, commits a parallel *cancellation process* P_{cancel} in which each sub-seller is compensated for early exit through a fee owed by the original buyer (and bonded under the standard kernel rule), and the new buyer commits a fresh transport process for the remaining legs.

Several open questions hold the pattern back from being a deliverable mechanism. First, the cancellable-seller wrapper as sketched in BoL Research [2026] relies on programmatic acknowledgement of the cancellation by the affected sub-seller under a pre-agreed fee schedule; whether this constitutes sub-seller consent in the kernel-discipline sense, or constitutes a unilateral exit dressed in counterparty clothing, is unsettled. A sub-seller bonded under a CancellableSeller wrapper has consented to the wrapper’s terms at commit time, but consent at commit time to a future programmatic acknowledgement is structurally different from consent at the moment of cancellation; the design discipline reading depends on which framing one accepts. Second, the cargo-handoff operationalization is unaddressed: a fresh transport process for the remaining legs requires the original carrier to release cargo to the new carrier, which is a real-world hand-off with no described attestation surface. Third, the cancellation-fee arithmetic is undefended; setting the fee too low produces an early-exit option the original sub-sellers will refuse, setting it too high forecloses the transfer use case, and the present paper does not defend any specific fee schedule.

The pattern is therefore not a recommended deployment artifact in the way the schemas of Section 6 are. We include the sketch because the BoL-transferability question is the load-bearing concern for any container-shipping deployment, and because the conventional answer (negotiable bills of lading endorsed mid-transit) is precisely what the kernel’s buyer-fixed invariant rules out. The honest position is that the pattern is open architectural research; deployments requiring BoL transferability should plan to rely on conventional BoL doctrine through the off-chain agreement until the pattern matures, or to constrain their use cases to the non-transferable region the kernel admits directly.

Why MLETR-style transferable-record analysis applies. The companion evidence-and-law paper develops a Class A / Class B records taxonomy that frames the MLETR analysis here precisely. *Class A* records are kernel byproducts (commitment digests, bond deposits, the cumulative-value accumulator state, resolution events): they are emitted by the kernel itself, schema-typed at the kernel level, and cryptographically authenticated. The MLETR Article 10–11 analysis applies directly to Class A records: kernel structural properties (singularity through the cumulative-value accumulator, control through buyer-dominance, integrity through the chain’s consensus) discharge the control and singularity tests for transferable records. *Class B* records — discretionary schema-validated attestations such as the `container-seal-v1` events or `handoff-v1` attestations of Section 4 — are evidentiary artifacts *about* the underlying transferable record (the commitment), not separate transferable records inheriting MLETR status in their own right. The cancellable-seller / counter-process pattern preserves the Class A fit because the transfer is achieved through bilateral signing on two separate processes rather than through unilateral endorsement of a single record; the protocol-layer transfer pattern does not violate either MLETR test.

What the bonded architecture does and does not preserve. A useful organizing distinction is between order-form (named consignee, transferred by endorsement to a designated party) and bearer-form (transferred by physical delivery of the instrument without endorsement) bills of lading. Order-form endorsement is partially expressible through the cancellable-seller pat-

tern above (subject to the open questions). Bearer-form BoLs are functionally equivalent to a transfer-without-counterparty-consent pattern that the bonded architecture does not support. Bearer-instrument practices remain common in commodity trading (oil cargoes, grain shipments) where the bill is treated as a near-money instrument [Goode and Gullifer, 2017]; non-bearer order-form practice predominates in container shipping but is not the only operational pattern. We treat the bearer-form exclusion as an honest exclusion: parties whose operations depend on bearer-instrument liquidity should plan around the conventional regime for that part of their workflow.

8 Comparison with Existing Approaches

The architecture should be located against the existing landscape of digital-shipping infrastructures.

TradeLens. TradeLens was a permissioned consortium ledger; the architecture proposed here is a permissionless bonded settlement primitive. The functional comparison is asymmetric: TradeLens supplied information transparency without settlement; the bonded architecture supplies settlement (with information transparency emerging as a byproduct of the bonded record). The governance comparison is also asymmetric: TradeLens was governed by IBM–Maersk; the bonded architecture has no governance at the kernel layer. The carriers who declined TradeLens did so for a governance reason; the architecture proposed here removes the reason.

CargoX. CargoX is a blockchain-based bill-of-lading platform that emphasizes transferable-electronic-record functionality and MLETR compliance [CargoX, 2023]. CargoX is a focused product addressing the BoL transferability problem; the bonded architecture is a more general settlement primitive whose relationship to CargoX is composable rather than competitive. A shipper using the bonded architecture for the bilateral transport coordination could compose CargoX (or its successor) at the BoL layer, with the bonded record providing the underlying settlement evidence and the CargoX layer providing the transferable-instrument semantics. The two are not in competition; they address adjacent layers of the stack.

TradeTrust. TradeTrust, a Singapore-government-backed framework for electronic transferable records, supplies open-source tools and standards for issuing and managing electronic BoLs and similar instruments [TradeTrust, 2023]. Like CargoX, it is composable with the bonded architecture rather than competitive with it. TradeTrust’s standards-based approach to MLETR fit produces artifacts that are interoperable with the bonded-architecture’s evidence record under the framework developed in the companion evidence-and-law paper.

MLETR. The UNCITRAL Model Law on Electronic Transferable Records (2017) is a legal framework rather than a technology platform [UNCITRAL, 2017]. The bonded architecture’s fit with MLETR is treated in the companion evidence-and-law paper. The short summary is that the kernel-layer artifacts satisfy Articles 10 and 11 (functional equivalence, control, singularity) for non-bearer transferable records, with bearer-instrument semantics noted as an honest exclusion (Section 7).

The architectural distinction. The TradeLens replacement is not a better TradeLens. It is a re-architecture of the underlying coordination function from *shared ledger maintained by consortium* to *bonded settlement primitive composed by participants*. The shift is structurally substantial: the consortium’s role is absorbed into the kernel’s permissionless protocol; the carriers’ participation decision is reduced to whether to compose with the protocol on individual

shipments rather than whether to ratify a competitor’s gatekeeping. The carriers who refused to ratify TradeLens are not asked to ratify anything in the bonded architecture; they participate by signing commitments to specific shipments and decline by not signing. The decision is made at the level of individual commerce rather than at the level of industry governance.

9 Scope

The argument bounds itself by several scope conditions.

Asset specificity at the port layer. Major ports are functional monopolies for the slot, terminal capacity, and operational infrastructure they provide; carriers have limited alternative-supplier leverage at congested hubs. The architecture’s competitive properties at the port-supply layer are constrained accordingly: where the port-of-call is dictated by the route and the port operator is a near-monopolist, the buyer-dominance leverage at the sub-procurement is attenuated. The architecture continues to function in these cases (the bond posture is well-defined; the resolution path is clean) but it does not produce price competition in port services where structural conditions don’t admit it.

Customs and sovereign authorities. Customs authorities are not Figaro counterparties; their decisions are sovereign and the architecture treats them as attestation sources rather than bonded parties. The architecture does not propose to re-engineer customs administration. It does provide a verifiable evidence record on which customs decisions can operate and which customs decisions are recorded against, which we take as an unambiguous improvement over the contemporary paper-and-database situation but not as a substitute for the sovereign authority itself.

Bond-currency at the small-shipper end. Small shippers without crypto-treasury infrastructure face operational friction at the bonding step. The constraint is real but solvable through standard treasury-of-record arrangements (a forwarder bonds on behalf of small shippers, with the small shipper paying conventional fees off-chain to the forwarder for the bonding service). This is a runtime-tier implementation question rather than a substrate-tier question; the substrate is denomination-agnostic.

Insurance and reinsurance. Marine insurance, P&I clubs, and the general apparatus of shipping-risk underwriting continue to operate. The bonded architecture changes the risk profile that the marine-insurance markets price (smaller residual after bond allocation; concentrated risk at the carrier-bond layer; a new class of bond-default risk at the port-supplier layer) but does not displace the insurance function. We expect the marine-insurance markets to develop fluent products around the bonded architecture as the architecture diffuses.

Network effects and adoption. The architecture’s adoption dynamics differ from TradeLens’s in that no participant must ratify any other participant’s governance position. A shipper, forwarder, or carrier can participate on individual shipments without joining anything; the participation decision is shipment-by-shipment rather than platform-membership-shaped. We do not model the diffusion trajectory beyond noting that the structural barrier that defeated TradeLens (consortium ratification by competitors) is absent here.

10 Conclusion

TradeLens failed for governance reasons that no consortium-governance fix can resolve, and the structural alternative is permissionless, ownerless infrastructure at the platform layer rather than a better consortium. The bonded settlement primitive of the companion mechanism paper supplies the architectural shape: each leg of a shipment is a bilateral bonded commitment; progressive collateralization scales the primitive across the multi-leg fan-out; buyer dominance plus atomic resolution propagates damage to the unit causing the slip; the schemas the assembly composes are largely existing infrastructure with two well-bounded additions.

The contribution is the assembly DAG, the per-edge mechanism, the bond-posture mechanics, the schema-set composition, the Incoterms-as-anchor pattern, and the cancellable-seller / counter-process pattern for BoL transferability. None of these is novel mechanism design; the mechanism-design work was done in the companion mechanism paper. The contribution is industrial-engineering: a specific re-architecture of container shipping as a bonded coordination problem with operational properties demonstrably different from the consortium-platform regime that has been tried and has failed.

A natural next step is empirical: a small-corridor pilot implementation of the architecture for a specific high-volume trade lane (US-Asia container, intra-EU short-sea, intra-Asia trade) with measurement of bond-loss frequency, port-supplier slip rates, and consignee-side recovery proportionality compared to the conventional regime. We do not undertake the pilot; we identify it as the experimental work the architecture invites.

Acknowledgments

The TradeLens worked example originated in a project repository of agent-coordination scenarios, with comparative research at BoL Research [2026]. The present paper formalizes the scenario into an industrial-engineering register.

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